A NOTE ON PROBLEM #650

1. Introduction

In [3], Erdős and Graham posed the following question, which also appears as Problem #650 on Bloom's Erdős Problems website [1].

Problem 1.1. *Let $f(m)$ be such that if $A \subseteq \{1, \dots, N\}$ has $|A| = m$ then every interval in $[1, \infty)$ of length $2N$ contains $\geq f(m)$ many distinct integers $b_1, \dots, b_r$ where each b_i is divisible by some $a_i \in A$, where $a_1, \dots, a_r$ are distinct.*

Estimate $f(m)$. In particular is it true that $f(m) \ll m^{1/2}$?

Erdős and Sarányi [4] proved that $f(m) \gg m^{1/2}$.

In this note, we solve Problem 1.1.

2. Main Result

Theorem 2.1. *For every integer $m \geq 1$, one has*

$$f(m) \leq 2\lceil \sqrt{m} \rceil.$$

More generally, for all integers $s, t \geq 1$,

$$f(st) \leq s + t.$$

In particular,

$$f(m) \ll \sqrt{m}.$$

Proof. We first prove the stronger statement

$$f(st) \leq s + t \qquad (s, t \geq 1).$$

Here, by a matching we mean a collection of distinct pairs (a_i, b_i) such that

$$a_i \in A, \qquad b_i \in I \cap \mathbb{Z}, \qquad a_i \mid b_i,$$

with all the a_i distinct and all the b_i distinct.

If $s = 1$ or $t = 1$, then any matching has size at most $|A| = st$, so trivially

$$f(st) \leq st \leq s + t.$$

Thus it remains to consider the case $s, t \geq 2$.

Define

$$E := \prod_{p \leq st} p^{1 + \lfloor \log_p(s-1) \rfloor}, \qquad D := E \cdot \operatorname{lcm}(1, 2, \dots, st).$$

Then for every prime $p \leq st$ and every integer δ with

$$1 \leq |\delta| \leq s - 1,$$

we have

$$v_p(D) > v_p(\delta),$$

where $v_p(\cdot)$ denotes the p-adic valuation. Indeed,

$$v_p(\delta) \leq \lfloor \log_p(s-1) \rfloor,$$

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while the exponent of p in E is

$$1 + \lfloor \log_p(s-1) \rfloor.$$

Next let

$$\mathcal{P} := \Big\{ p > st : \ p \text{ is prime and } p \mid (qD + \delta) \text{ for some } 1 \le q \le t-1, \ 1 \le |\delta| \le s-1 \Big\}.$$

This is a finite set of primes.

For each $p \in \mathcal{P}$, consider the residue classes

$$i - jD \pmod p \qquad (1 \le i \le s, \ 0 \le j \le t-1).$$

There are at most st such classes, whereas $p > st$, so they do not exhaust all residue classes modulo p. Hence we may choose a residue class

$$r_p \pmod p$$

such that

$$r_p \not\equiv i - jD \pmod p \qquad (1 \le i \le s, \ 0 \le j \le t-1).$$

By the Chinese remainder theorem, there exists an integer a such that

$$a \equiv r_p \pmod p \qquad (\forall p \in \mathcal{P}),$$

and also

$$a > 2(t-1)D + 4s.$$

For $1 \le i \le s$ and $0 \le j \le t-1$, define

$$a_{i,j} := a + jD - i.$$

These integers are pairwise distinct. Indeed, if

$$a + jD - i = a + \ell D - k,$$

then

$$(j - \ell)D = i - k.$$

Since

$$|i - k| < s \le st \le D,$$

it follows that $j = \ell$, and then $i = k$.

Now set

$$A := \{a_{i,j} : 1 \le i \le s, \ 0 \le j \le t-1\}, \qquad N := a + (t-1)D - 1.$$

Then $|A| = st$, and every element of A lies in $\{1, \dots, N\}$, because

$$1 \le a - s \le a_{i,j} \le a + (t-1)D - 1 = N.$$

We claim that for any two pairs (i,j) and (k,ℓ),

$$\gcd(a_{i,j}, a_{k,\ell}) \mid (k - i).$$

Let

$$d := \gcd(a_{i,j}, a_{k,\ell}).$$

Since

$$a_{i,j} - a_{k,\ell} = (j - \ell)D + (k - i),$$

we have

$$d \mid (j - \ell)D + (k - i).$$

We distinguish three cases.

Case 1: $i = k$.

Then

$$d \mid 0 = (k - i),$$

so there is nothing to prove.

Case 2: $i \neq k$ and $j = \ell$.

Then immediately

$$d \mid (k - i).$$

Case 3: $i \neq k$ and $j \neq \ell$.

Let

$$q := |j - \ell| \in \{1, \dots, t-1\},$$

and choose

$$\delta = \begin{cases} k - i, & j > \ell, \\ i - k, & j < \ell. \end{cases}$$

Then

$$d \mid qD + \delta, \qquad 1 \leq |\delta| = |k - i| \leq s - 1.$$

We first show that every prime divisor of d is at most st. Suppose, for contradiction, that some prime $p > st$ divides d. Since

$$p \mid d \mid qD + \delta,$$

we have $p \in \mathcal{P}$ by definition of $\mathcal{P}$. On the other hand,

$$p \mid d \mid a_{i,j} = a + jD - i,$$

so

$$a \equiv i - jD \pmod{p},$$

contradicting the choice of $a \equiv r_p \pmod{p}$, where r_p avoids all residue classes $i - jD \pmod{p}$. Thus every prime divisor of d is at most st.

Now let $p \leq st$ be any prime. Since

$$v_p(qD) \geq v_p(D) > v_p(\delta),$$

it follows that

$$v_p(qD + \delta) = v_p(\delta).$$

Because $d \mid qD + \delta$, we obtain

$$v_p(d) \leq v_p(qD + \delta) = v_p(\delta) = v_p(k - i).$$

Since all prime divisors of d are at most st, we conclude that

$$d \mid (k - i).$$

This proves the claim.

Consequently, the system of congruences

$$x \equiv -i \pmod{a_{i,j}} \qquad (1 \leq i \leq s,\ 0 \leq j \leq t-1)$$

is pairwise compatible: for any two congruences in the system, the difference of the residues is

$$(-i) - (-k) = k - i,$$

and by the claim this is divisible by $\gcd(a_{i,j}, a_{k,\ell})$. Therefore, by the generalized Chinese remainder theorem [2, Theorem 11.5], the system has a solution x_0.

Replacing x_0 by a sufficiently large positive integer in the same residue class if necessary, we may assume that the interval introduced below lies in $[1, \infty)$.

Choose an integer T such that

$$2\big((t-1)D + s - 1\big) < T \leq a - 2s.$$

Such a choice is possible because

$$a > 2(t-1)D + 4s.$$

Consider the interval

$$I := (x_0 - T,\ x_0 - T + 2N].$$

Its length is $2N$.

Fix (i, j) with $1 \le i \le s$ and $0 \le j \le t - 1$. Since

$$x_0 \equiv -i \pmod{a_{i,j}},$$

we have

$$a_{i,j} \mid (x_0 + i).$$

Also, since

$$a_{i,j} = a + jD - i,$$

we have

$$x_0 + a + jD = (x_0 + i) + a_{i,j},$$

so $x_0 + a + jD$ is another multiple of $a_{i,j}$.

We now show that these are exactly the two multiples of $a_{i,j}$ contained in I.

First, both of them lie in I. Indeed, the offset of $x_0 + i$ from the left endpoint $x_0 - T$ is

$$T + i,$$

which is positive and satisfies

$$T + i \le T + s \le a - s < 2N.$$

Hence

$$x_0 + i \in I.$$

Similarly, the offset of $x_0 + a + jD$ from the left endpoint is

$$T + a + jD,$$

which is again positive. Moreover,

$$T + a + jD \le T + a + (t - 1)D \le (a - 2s) + a + (t - 1)D < 2N,$$

since

$$2N = 2a + 2(t - 1)D - 2$$

and

$$(a - 2s) + a + (t - 1)D < 2a + 2(t - 1)D - 2$$

for $s, t \ge 2$. Thus

$$x_0 + a + jD \in I.$$

Next, the preceding multiple of $a_{i,j}$ is

$$(x_0 + i) - a_{i,j}.$$

Its offset from the left endpoint is

$$T + i - a_{i,j}.$$

But

$$a_{i,j} = a + jD - i \ge a - s,$$

and

$$T + i \le T + s \le a - s \le a_{i,j},$$

so

$$T + i - a_{i,j} \le 0.$$

Hence this preceding multiple is not contained in I.

Finally, the next multiple after $x_0 + a + jD$ is

$$(x_0 + i) + 2a_{i,j},$$

whose offset from the left endpoint is

$$T + i + 2a_{i,j} \geq T + 1 + 2(a - s).$$

Using

$$T > 2\big((t-1)D + s - 1\big) = 2(t-1)D + 2s - 2,$$

we obtain

$$T + i + 2a_{i,j} > 2(t-1)D + 2s - 2 + 1 + 2a - 2s = 2a + 2(t-1)D - 1 > 2N,$$

because

$$2N = 2a + 2(t-1)D - 2.$$

Therefore this multiple also lies outside I.

Thus $a_{i,j}$ has exactly two multiples in I, namely

$$x_0 + i \qquad \text{and} \qquad x_0 + a + jD.$$

It follows that every integer in I divisible by some element of A lies in the set

$$B := \{x_0 + i : 1 \leq i \leq s\} \cup \{x_0 + a + jD : 0 \leq j \leq t-1\}.$$

The two parts of this union are disjoint, since

$$x_0 + i = x_0 + a + jD$$

would imply

$$i = a + jD,$$

which is impossible because $1 \leq i \leq s < a \leq a + jD$. Hence

$$|B| = s + t.$$

Therefore, inside the interval I, there are at most $s + t$ integers divisible by some element of A. In particular, there cannot exist a matching from A into $I \cap \mathbb{Z}$ of size larger than $s + t$. Hence

$$f(st) \leq s + t.$$

Now let

$$n := \lceil \sqrt{m} \rceil.$$

Applying the above result with $s = t = n$, we obtain a set

$$A \subseteq \{1, \ldots, N\}$$

of size n^2 and an interval I of length $2N$ such that at most $2n$ integers in I are divisible by some element of A.

Choose any subset $A' \subseteq A$ with $|A'| = m$. Then $A' \subseteq \{1, \ldots, N\}$, and the set of integers in I divisible by some element of A' is contained in the corresponding set for A, so it still has cardinality at most $2n$. Therefore there cannot exist a matching from A' into $I \cap \mathbb{Z}$ of size larger than $2n$, and hence

$$f(m) \leq 2n = 2\lceil \sqrt{m} \rceil.$$

This completes the proof. □

References

[1] T. F. Bloom, Erdős Problem #650, `https://www.erdosproblems.com/650`, accessed 2026-03-06.

[2] N. Donaldson, *Math 180A - Notes*, University of California, Irvine, March 14, 2018. Available at: `https://www.math.uci.edu/~ndonalds/math180a/notes.pdf` (accessed March 7, 2026).

[3] P. Erdős, Some problems in number theory. Octogon Math. Mag. (1995), 3–5.

[4] Erdős, P. and Sarányi, Megjegyzések egy versenyfeladathoz. Matematikai Lapok (1959).